Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Physical-AI Augmented Reality Glasses for SLAM-Enabled Real-Time Operator Assistance in Aerospace Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical - Materials		*Mechanical - Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Aerospace Systems	*Aerospace System Software*	*Robotics/Automation Machinery*	*Measurement/Sensor Technologies for Automation*
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The proposed technology is a wearable Physical-AI AR glasses platform that provides real-time, hands-free guidance for aerospace and marine manufacturing operators performing complex, safety-critical tasks. Using SLAM-based spatial intelligence (Simultaneous Localization and Mapping), the system continuously localizes the operator and maps the workspace to deliver stable, accurate overlays in dynamic shop-floor conditions. It projects context-aware visual cues, step-by-step procedures, quality checkpoints, and hazard prompts directly into the operator's field of view to improve execution while minimizing distraction. Key functions include: • Workflow guidance and adaptive task support • Real-time hazard identification and alerting • Automated logging and performance analytics • Integration with enterprise and sensor systems for remote assistance, reporting, and traceability Target applications include aerospace/marine assembly and inspection, maintenance and repair operations (MRO), logistics and kitting verification, and training in environments where in-situ decision support improves safety, quality, and productivity.			

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Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* Operators in dynamic industrial environments are required to perform complex tasks while simultaneously managing multitasking demands, real-time decision-making, and strict safety requirements. In aerospace and marine manufacturing and maintenance, work is often conducted under time pressure, physical constraints, and high cognitive load, where even minor errors can lead to safety incidents, quality issues, or costly rework [1-4]. Although AR technologies have shown potential for supporting operators [5], current solutions remain limited in their adaptability and practical deployment. Many AR systems are highly task-specific, difficult to customize across different operations, and constrained by limited access to on-board sensors that can be seamlessly integrated into a unified development and validation environment. As a result, there is a clear gap between AR concepts demonstrated in laboratory settings and robust, deployable systems suitable for real-world industrial use [6-8]. This project addresses the gap by introducing a Physical AI approach, where intelligence is embodied in a wearable system that can perceive, localize, and adapt within real operational spaces. In particular, SLAM (Simultaneous Localization and Mapping) enables AR glasses to continuously map aerospace workstations and maintain stable, accurate overlays despite environmental changes, motion, and occlusions—capabilities that are essential for field-ready deployment [9-11]. As a Professor of Aerospace Engineering, Dr. Joon Chung has led numerous high-profile collaborative projects at the interface of academia and industry: • Holodeck Project (Bombardier-led): Developed AR-enabled Electronic Flight Instrument Systems (EFIS) for real-time cockpit interface prototyping and evaluation. • AI-enabled Metaverse Campus (OVA-led, Canada's Digital Supercluster): Introduced haptic gloves and immersive training checklists for Canadair CL-515 operations. • Advanced Airspace Usability (ADAIR, Bombardier-led consortium): Development of conformal HUD symbologies for trajectory-based operations, addressing cockpit decision support. • Mitacs–Columbiad Projects (multiple IT-series grants): XR-enabled control stations with multi-user collaboration, VR, and haptics for remote operation of UAVs, and space applications. A joint R&D model is required because no single organization can efficiently deliver both (1) the research-grade AR/AI methods and validation rigor and (2) the manufacturable, sensor-ready wearable platform needed for commercialization. Dr. Chung's lab will lead AR/AI architecture, spatial intelligence algorithms, human-centered interaction design, and system-level optimization, and will independently validate performance through rigorous experimental protocols and KPI-based evaluation. The Korean SME partner will lead wearable hardware prototyping, sensor integration, embedded optimization (power/thermal/connectivity), and industrial system interfacing required for robust deployment. Collaboration is therefore essential to ensure the solution is technically robust, validated under realistic conditions, and aligned with commercial constraints such as wearability, efficiency, and manufacturability. By tightly coupling academic research with industrial development, this joint R&D will move beyond isolated prototypes toward a deployable, scalable platform with clear pathways to industrial adoption and commercialization.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Physical AI 2. Augmented Reality Glasses 3. Real-Time Operator Assistance 4. Aerospace Manufacturing 5. SLAM (Simultaneous Localization and Mapping)

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* This joint R&D project will develop an integrated Physical-AI AR-glasses operator-assistance platform for aerospace and marine manufacturing and maintenance, advancing the technology from TRL 2 to TRL 5 through coordinated hardware development, AI-enabled perception and guidance, and system-level validation in representative environments. First, the project will design and prototype ergonomic, optical see-through AR glasses optimized for shop-floor use. The wearable hardware will support high-fidelity, low-latency visual overlays, including step-by-step procedures, quality checkpoints, hazard cues, and in-situ annotations, while ensuring comfort, stability, and safe hands-free operation under aerospace operational constraints (PPE compatibility, confined workspaces, variable lighting, and noise). Second, the project will implement the core AR perception and software stack required for robust deployment. This includes industrial sensor and asset connectivity middleware, a workflow authoring toolchain, and, critically, SLAM-based spatial intelligence (simultaneous localization and mapping) to continuously localize the operator and map the workspace. SLAM will enable accurate spatial registration and persistent overlay stability despite motion, occlusions, and changing shop-floor conditions, which is essential for aerospace assembly, inspection, and MRO tasks. Third, the project will develop AI-driven real-time guidance and contextual assistance at the decision layer. The AI modules will fuse multi-source context—operator interaction patterns, task progression, tool/part identification, spatial location, and sensor signals—to infer latent operational states not captured by predefined rules alone. Based on this inference, the system will deliver adaptive guidance with appropriate timing and priority, including next-best action recommendations, anomaly and hazard detection, quality assurance prompting, and escalation to remote experts under uncertainty. AI models will be trained on representative aerospace workflows and optimized for low-latency edge inference to maintain responsiveness in real industrial conditions. Finally, the integrated platform will be validated under realistic operational scenarios using quantitative KPIs such as task completion time, first-pass quality/accuracy, error rate reduction, safety compliance, and usability. System-level optimization will address power consumption, thermal behavior, connectivity robustness, and logging/traceability to support reliable operation across extended work periods, providing objective evidence of TRL 5 readiness and commercialization potential.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* The target performance specifications for the TRL 5 prototype are defined to ensure reliable, safe, and high-quality operation in demanding aerospace and marine manufacturing/MRO environments, where variable lighting, occlusions, confined workspaces, PPE constraints, and continuous task execution require robust wearable performance. **AR Display and Visual Performance** • Optical see-through AR overlays with ≥2K effective resolution per eye (≈2,000×1,000) • ≥40° diagonal field of view (FOV) to support in-situ guidance without excessive head motion • ≥500 nits overlay brightness (target ≥1,000 nits) ensuring legibility under 0–50,000 lux ambient lighting • Overlay readability under reflective metallic surfaces common in aerospace production environments (qualitative pass/fail criterion)			

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	SLAM-Based Spatial Intelligence and Registration Performance • SLAM-enabled localization and mapping stability sufficient for dynamic shop-floor conditions • ≤10 mm spatial registration error at 1 m working distance (target ≤5 mm) • Full 6DoF tracking with persistent spatial anchoring across operator movement and viewpoint changes • Positional drift ≤1% over a 10-minute task and re-localization recovery within ≤2 seconds after brief occlusion or tracking loss **System Latency and Rendering Performance** • End-to-end motion-to-photon latency ≤50 ms (target ≤35 ms) for stable overlays during motion • Stable rendering performance ≥60 fps (target ≥72 fps) to minimize discomfort and support precision tasks • Overlay jitter maintained below operator-noticeable thresholds during typical head motion (validated experimentally) **AI Guidance and Real-Time Responsiveness** • Contextual state inference and guidance updates at ≥5 Hz (target ≥10 Hz) • System response time to operator prompts ≤500 ms (target ≤300 ms) • Anomaly/hazard prompt triggering within ≤1 second from detection to display under representative conditions **Wearability and Ergonomics** • Total device mass ≤350 g (excluding battery) with balanced weight distribution • Continuous operation time ≥3 hours (target ≥4 hours) with hot-swappable battery support • Maximum temperature of skin-contact surfaces (temple arms, nose bridge, forehead-contact pads): ≤42°C during continuous operation • Maximum temperature of non-contact external surfaces: ≤48°C during continuous operation, with no hot spots accessible during normal wear and handling • Comfortable use with standard PPE (safety glasses/helmet compatibility) validated through user trials **Robustness and Interaction Performance** • Environmental protection rating IP54 (dust/splash resistance) • Reliable operation across 0–40°C ambient temperature range • ≥95% accuracy for voice and gesture command recognition under noise levels up to 80 dB • Network resilience: maintain core guidance functions under intermittent connectivity, with offline caching and automatic resynchronization Together, these quantitative performance targets define the technical baseline required to demonstrate TRL 5 readiness, ensuring dependable operation, stable SLAM-based overlays, and safe real-time assistance under representative aerospace industrial conditions.
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* **R&D Plan (36 months, TRL 2 → TRL 5)** **Approach** This project will follow an iterative, user-centered, model-based R&D approach tailored to aerospace and marine manufacturing/MRO environments, and tightly couples: (i) Wearable AR hardware/optics and ergonomics, (ii) Physical-AI perception enabled by SLAM-based spatial intelligence for robust localization and mapping, and (iii) AI-driven contextual decision-making as the guidance layer that adapts procedures in real time.

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Progression from TRL 2 to TRL 5 will be achieved through increasingly realistic demonstrations, evolving from concept validation, to integrated laboratory prototypes, and finally to pilot validation in representative operational settings. Each phase includes explicit Physical AI performance targets (SLAM stability, spatial accuracy, decision latency, and robustness under occlusions) and formal verification milestones to support TRL advancement.

Phase 1 — Foundations and TRL 2→3 (Months 1–12)

Task 1. Requirements and Use-Case Definition (M1–M3)

Select 2–3 representative aerospace/marine tasks requiring high precision and safety (e.g., assembly/fastening sequences, inspection checklists, kitting/logistics verification, line maintenance steps). Define measurable KPIs including task completion time, first-pass quality/accuracy, error rate and rework, near-miss events, safety compliance, cognitive workload, usability (e.g., SUS), and system reliability. In parallel, define AI-specific performance requirements including decision latency, robustness to task variability, acceptable false-alarm/miss rates, and operator trust criteria. Define SLAM-related requirements for localization continuity, relocalization time, and overlay stability in shop-floor conditions.

Task 2. System Architecture and Interfaces (M1–M4)

Specify the end-to-end architecture including: industrial sensor and enterprise ingestion (tool/asset identifiers, work orders, equipment states), SLAM-based tracking and spatial registration, overlay rendering engine, AI guidance and decision modules, and remote expert support. Explicitly locate AI at the decision layer and SLAM at the perception layer, with defined APIs, data formats, logging schemas, cybersecurity considerations, and interface control documents (ICDs). This ensures seamless integration between wearable device firmware, AR software, AI inference modules, and industrial systems.

Task 3. Early Hardware and Ergonomic Proof-of-Concept (M3–M10)

Develop early wearable AR hardware prototypes and integration configurations, including optics/display, compute, battery, thermal subsystems, cameras/IMU depth/vision sensors (as applicable), and connectivity modules. Conduct iterative fit and comfort evaluations to assess wearability, stability, field-of-view usability, interaction suitability, and PPE compatibility. Hardware constraints relevant to real-time AI inference (compute headroom, power draw, thermal margins) will be quantitatively assessed and used to guide later optimization and edge deployment.

Task 4. Physical-AI AR Pipeline Proof-of-Concept (M4–M12)

Implement an initial AR pipeline featuring calibration, spatial anchoring, overlay rendering, and a preliminary workflow authoring tool. Develop SLAM proof-of-concept capabilities to demonstrate robust 6DoF localization and overlay stability during controlled motion and partial occlusions. Validate baseline guidance logic in controlled settings with early datasets, including timing behavior (guidance responsiveness) and fail-safe behavior under uncertainty.

Milestones:

- **M6**: Requirements baselined; architecture and ICDs frozen
- **M12**: TRL 3 demonstration achieved (stable overlays + early SLAM-based anchoring + basic AI-assisted guidance in controlled lab setup)

Phase 2 — Integrated prototype and TRL 3→4 (Months 13–24)

Task 5. Sensor Integration, Data Infrastructure, and Robustness (M13–M20)

Integrate selected on-board sensors and relevant industrial data sources via standardized middleware. Implement time synchronization, confidence scoring, uncertainty estimation, and fault handling for missing/noisy signals. Establish a structured data pipeline (logs, events, guidance actions, operator interactions) to enable AI training and validation while supporting traceability required in aerospace manufacturing contexts.

Task 6. AI-Driven Contextual Guidance and Decision Support (M13–M22)

Develop AI modules operating at the decision layer: step recommendation, task progression inference, anomaly and deviation detection, and decision support under uncertainty. Methodologies include dataset specification, ground-truth labeling, offline model training, and

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edge-optimized inference for low-latency operation. Continuous improvement loops will be supported through structured pilot logs. Human-in-the-loop mechanisms and rule-based safety constraints will be embedded to ensure predictable, explainable, and safe behavior (e.g., conservative alerts, escalation when confidence drops, and operator override).

Task 7. Human Factors, Interaction Design, and Safety-Oriented UI (M13–M24)

Refine interaction modalities (voice, gesture, physical controls) and information presentation strategies (alert timing, priority cues, minimal text, spatial cueing) to reduce distraction and support situational awareness. Conduct controlled experiments to evaluate operator workload, error recovery, usability, trust in AI prompts, and the effect of SLAM overlay stability on task performance. Iterate UI layouts and interaction policies based on quantitative metrics and operator feedback.

Milestones:

- **M18**: Integrated prototype v1 with sensor feeds + SLAM-based anchoring + AI guidance
- **M24**: TRL 4 validation completed in laboratory/bench environments with quantified KPI improvements vs. baseline procedures

Phase 3 — Representative validation and TRL 4→5 (Months 25–36)

Task 8. Pre-Pilot Hardening and Edge Optimization (M25–M30)

Optimize system performance for representative deployment: improve AI inference latency, reduce power consumption, stabilize thermal behavior (skin-contact vs. non-contact surface limits), and enhance network resilience and offline continuity. Strengthen SLAM robustness for industrial conditions (relocalization time, drift control, occlusion recovery). Harden the workflow authoring toolchain to enable rapid updates of procedures and content without full retraining of core AI models, allowing adaptation across different aerospace tasks and sites.

Task 9. Representative Operational Trials and KPI Benchmarking (M28–M35)

Execute pilot evaluations in task-representative environments with realistic layout variability, lighting, reflective surfaces, noise, motion, and PPE constraints. Benchmark against baseline SOPs (paper/tablet-based guidance and conventional inspection practices). Collect quantitative evidence including completion time, first-pass quality, error and rework rates, safety compliance, and operator workload. Evaluate SLAM performance (registration accuracy, drift, relocalization) and AI performance (false alarms/misses, timing, decision utility) with statistical analysis.

Task 10. TRL 5 Demonstration Package and Commercial Readiness Outputs (M33–M36)

Deliver a complete TRL 5 demonstrator, comprising AR glasses hardware, guidance software, sensor/enterprise integration middleware, validated workflows, and an evaluation report evidencing performance gains and robustness. Produce deployment guidelines covering training, safety, calibration routines, device maintenance, and AI performance monitoring (drift detection, update strategy, and confidence reporting). Outputs will be structured to enable commercialization planning and early adopter onboarding.

Milestones:

- **M30**: Pilot-ready prototype v2 (hardened SLAM + optimized edge inference)
- **M36**: TRL 5 demonstration achieved, supported by repeatable performance evidence and a commercialization-ready technical package

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* The Research Institute, University of Toronto (U of T), will lead the scientific, methodological, development, and validation components of the project to advance the Physical-AI AR operator-assistance platform from TRL 2 to TRL 5. U of T's overall objective is to define, develop, and independently validate a robust AR/SLAM/AI capability stack that delivers reliable, safe, and measurable performance improvements in aerospace and marine manufacturing/MRO environments, with clear pathways to commercial deployment.

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	U of T will serve as the technical and system-level lead, responsible for end-to-end architecture definition, AR/AI software development, SLAM-based spatial intelligence integration, system verification, and evidence-based TRL progression. U of T will ensure that the platform is designed using a user-centered approach and validated through rigorous experimental protocols under realistic operational constraints (PPE, lighting variability, occlusions, noise, and constrained workspaces). Specifically, U of T will: 1. Lead requirements engineering and aerospace use-case design, translating operational workflows into measurable task models, KPIs, and acceptance criteria (e.g., task time, first-pass quality, error/rework rate, safety compliance, usability, and system reliability). 2. Develop the core AR perception and software pipeline, including calibration, overlay stability, workflow authoring methods, and integration of SLAM-based localization and mapping for accurate spatial anchoring and persistent overlays in dynamic environments. 3. Design and implement AI-driven contextual assistance, including multi-source context fusion, decision support, anomaly and hazard prompting, confidence-based escalation, and human-in-the-loop safeguards to ensure predictable behavior, transparency, and operator trust. 4. Plan and execute system-level verification and validation, including experimental protocols, benchmarking against baseline procedures (paper/tablet SOPs), statistical performance analysis, and TRL evidence documentation from 2 to 5. 5. Provide research resources and infrastructure, including faculty supervision, postdoctoral researchers, graduate-level trainees, AR simulation/prototyping toolchains, software development environments, and evaluation facilities for rapid iteration and repeatable testing. 6. Lead human-centered evaluation, including usability testing, cognitive workload assessment, interaction design refinement, and safety-oriented information presentation to minimize distraction and maximize situational awareness. 7. Support system optimization, including latency budgeting, edge-inference performance tuning, SLAM robustness enhancement, and reliability improvements under variable environmental and connectivity conditions. In addition to technical leadership, U of T offers a strong cost and commercialization advantage. Compared with typical R&D costs in the USA and Europe, Canada can provide approximately one-third lower cost for hiring graduate-level researchers, enabling deeper R&D execution within the same budget. U of T also provides industry-friendly IP practices and a lower overhead model (≈40% of direct costs), improving budget efficiency and accelerating translation to commercial prototypes. Finally, U of T will support IP strategy and dissemination through technical reports, TRL evidence packages, evaluation datasets (where permissible), and deployment guidelines aligned with partner productization objectives.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* The Korean SME partner will lead the hardware engineering, embedded implementation, system integration, and deployment-readiness activities required to advance the solution from early prototypes to a TRL 5, pilot-ready Physical-AI AR-glasses platform for aerospace and marine manufacturing/MRO. The SME's core responsibility is to ensure the wearable device and its embedded software stack are manufacturable, reliable, and deployable in representative operational environments, while collaborating closely with the Research Institute to align hardware capabilities with the AR/SLAM/AI performance requirements. First, the SME will manufacture AR hardware prototypes through iterative development cycles across the project. This includes optical see-through display modules, onboard compute, cameras/IMU and tracking sensors, power and battery subsystem (including hot-swappable configurations where applicable), thermal management solutions, mechanical enclosure design, and ergonomic fit optimization to ensure comfort, stability, and safety during extended wear in

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	industrial conditions. The SME will manage component selection, supplier coordination, assembly workflows, and quality checks to de-risk scalability beyond laboratory prototypes and to support future productization. Second, the SME will enable SLAM-ready sensing and real-time operation by supporting sensor calibration procedures, synchronized data acquisition, and embedded performance tuning necessary for robust localization and mapping on a wearable platform. This includes optimizing compute utilization, thermal margins, and power efficiency to maintain stable AR overlays and low-latency execution while operating continuously over extended work periods. Third, the SME will integrate the platform with industrial systems by implementing and maintaining connectivity to shop-floor and operational data sources such as asset identifiers, work-order systems, equipment state signals, and edge gateways. This scope includes firmware and driver development, interface control and API alignment, device provisioning and management, secure communication protocols, and performance tuning to ensure dependable, low-latency operation under variable network conditions and real-world deployment constraints. Fourth, the SME will support system validation and troubleshooting by providing engineering staff, prototype test equipment, and integration support during laboratory evaluations and representative-environment trials. The SME will diagnose hardware–software integration issues and harden the platform for operational robustness, including fault recovery, calibration stability, relocalization recovery support, network resilience, and tolerance to environmental conditions (lighting variability, noise, temperature, dust/splash exposure). The SME will also contribute deployment logistics to enable repeatable test campaigns and rapid iteration cycles. Finally, the SME will generate technical documentation required for TRL evidence and commercialization readiness, including hardware specifications, integration/interface manuals, calibration and maintenance procedures, and deployment checklists. The SME will actively participate in joint design reviews, sprint planning, and risk management to ensure co-development decisions remain aligned with measurable KPI targets, aerospace deployment constraints, and commercialization objectives.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* This joint R&D project will deliver a TRL 5 Physical-AI AR-glasses operator-assistance platform that integrates ergonomic, optical see-through wearable hardware with a robust AR/SLAM/AI software stack to provide real-time guidance in demanding aerospace and marine industrial environments. The primary technological outcome is the development of wearable AR glasses optimized for prolonged use, capable of presenting high-fidelity overlays—procedural steps, quality checkpoints, hazard cues, and in-situ annotations—through intuitive, hands-free interaction while minimizing distraction and supporting safe execution. A second major outcome is the realization of SLAM-based spatial intelligence on the wearable device, enabling continuous localization and workspace mapping to maintain stable overlays under real shop-floor conditions (operator motion, occlusions, reflective surfaces, and layout variability). This capability addresses a key barrier to practical AR deployment by ensuring reliable spatial alignment between digital instructions and the physical workspace, which is essential for aerospace assembly, inspection, and maintenance operations where precision and traceability are critical. In parallel, the project will deliver AI-driven contextual assistance that fuses operational context and sensor inputs to recommend next-best actions, detect deviations from expected workflows, and issue confidence-aware hazard prompts. When uncertainty is high, the system will support escalation to remote experts and enable structured logging for post-task review. Together, these technologies establish a validated, system-level solution suitable for safety-critical industrial applications where errors, rework, and incidents carry significant operational and economic cost. A core outcome will be a pilot-validated demonstrator evaluated in representative aerospace

	manufacturing workflows. Quantitative evidence will be generated to demonstrate performance improvements over baseline procedures, including reduced task completion time, improved first-pass quality, and lower error rates. By validating repeatable performance under realistic constraints—variable lighting, high noise, PPE usage, operator motion, and intermittent connectivity—the project will deliver a commercialization-ready engineering package, including integration middleware, workflow authoring methods, validation reports, and deployment guidelines. From a commercialization perspective, the project positions the Korean industrial partner to accelerate productization and market entry with a differentiated operator-support solution aligned with enterprise needs in aerospace manufacturing and adjacent sectors (marine systems, heavy industry, maintenance, and inspection). Expected benefits include reduced operational errors and rework, enhanced safety compliance, faster onboarding and training, and measurable cost savings from reduced downtime and improved throughput. Beyond individual deployments, the project will establish a scalable development model—combining “sensor-to-guidance” integration and reusable content-authoring—enabling rapid adaptation to new tasks and sites. At a broader level, the project supports Korea’s strategic priorities in AI-enabled industrial systems, advanced robotics, and aerospace/marine manufacturing, strengthening competitiveness and export potential.

6. References

(1) M. Rostami, P. Pradhan, N. Karki, J. Omorodion, P. Milani, J. Kamoonpuri, H. Y. Liu, J. Chung, “A comprehensive review of extended reality and its application in aerospace engineering,” Progress in Aerospace Sciences, Vol. 157, 2025, 101118, ISSN 0376-0421, https://doi.org/10.1016/j.paerosci.2025.101118.
(2) A. K. Gramopadhye and C. G. Drury, “Human factors in aviation maintenance: how we got to where we are,” Int. J. Ind. Ergon., vol. 26, no. 2, pp. 125–131, Aug. 2000, doi: 10.1016/S0169-8141(99)00062-1.
(3) S. Dalkilic, “Improving aircraft safety and reliability by aircraft maintenance technician training,” Eng. Fail. Anal., vol. 82, pp. 687–694, Dec. 2017, doi: 10.1016/j.engfailanal.2017.06.008.
(4) Y.-H. Chang and Y.-C. Wang, “Significant human risk factors in aircraft maintenance technicians,” Saf. Sci., vol. 48, no. 1, pp. 54–62, 2010.
(5) K. Kim, J. Chung, W. Chapman, J. M. Kamoonpuri, B. Gwak, A. Choe, M. Rostami, H. Y. Liu, P. Pradhan, “Extended Reality Communications Environment,” U.S. Patent 12,400,408, 2025, https://patents.google.com/patent/US12400408B2/en
(6) A. Y. C. Nee, S. K. Ong, G. Chryssolouris, and D. Mourtzis, “Augmented reality applications in design and manufacturing,” CIRP Ann., vol. 61, no. 2, pp. 657–679, Jan. 2012, doi: 10.1016/j.cirp.2012.05.010
(7) E. Bottani, and G. Vignali, “Augmented reality technology in the manufacturing industry: A review of the last decade,” IISE Transactions, vol. 51, no.3, pp. 284–310, 2019, https://doi.org/10.1080/24725854.2018.1493244
(8) R. Palmarini, J. A. Erkoyuncu, R. Roy, and H. Torabmostaedi, “A systematic review of augmented reality applications in maintenance,” Robot. Comput-Integr. Manuf., vol. 49, pp. 215–228, Feb. 2018, doi: 10.1016/j.rcim.2017.06.002
(9) P. E. Sarlin, M. Dusmanu, J. L. S., P. Speciale, L. Gruber, V. Larsson, O. Miksik, and M. Pollefeys, “LaMAR:Benchmarking Localization and Mapping for Augmented Reality,” Computer Vision – ECCV 2022. ECCV 2022. Lecture Notes in Computer Science, vol 13667. Springer, Cham. https://doi.org/10.1007/978-3-031-20071-7_40
(10) W. Fang, L.Chen, T. Zhang, C. Chen, Z. Teng, L. Wang, “Head-mounted display augmented reality in manufacturing: A systematic review,” Robotics and Computer-Integrated Manufacturing, vol. 83, 2023, 102567, https://doi.org/10.1016/j.rcim.2023.102567.
(11) K. Kurzhals, M. Becher, N. Pathmanathan and G. Reina, "Evaluating Situated Visualization in AR with Eye Tracking," 2022 IEEE Evaluation and Beyond - Methodological Approaches for Visualization (BELIV), Oklahoma City, OK, USA, 2022, pp. 77-84, doi: 10.1109/BELIV57783.2022.00013.

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization